Computer Networks

Midterm – Pattern

Prepare lectures covered in slides 1,2,3

Total Marks: 30
Total Time: 1 hour

Final Exam Pattern

Prepare lectures covered in slides

- CN - Slides 2.pdf
- CN - Slides 3.pdf
- CN - Slides 4.pdf
- CN - Slides 5.pdf
- CN - Slides 6.pdf
- CN - Slides 7.pdf

Topics to Prepare:

- Data link layer functionality, MAC addressing (unicast,
- multicast, broadcast) , CSMA, Types of Transmission (Asynchronous and Synchronous Transmission)
- Multiplexing:
- Networking devices, Network layer protocols (IPv4 and IPv6)
- Transport layer protocols, ports and sockets, TCP and UDP
- Application layer protocols
- Wireless Networks standards and technologies
- Processes communication, Sockets, Web and HTTP, Web Caches, DNS, Multiplexing, demultiplexing, TCP, UDP, Congestion Control, Flow Control, SDN, network service model, packet scheduling policies, IP addressing, Subnets, DHCP, NAT, SNMP, CSMA/CD, MAC Addressing, Switch, Router, Ethernet, VPN, Wireless and Wired Networks

Types of Questions:

4 questions
Q1. Theoretical- 5 Marks [8 minutes]

3 questions:
Q1. Conceptual + diagram – 10 marks {algorithm} [20 minutes]
Q2. Conceptual + diagram – 10 Marks [20 minutes]
Q3. Conceptual + diagram – 10 Marks [20 minutes]

Sample Question:
Part A:
What problem does DNS solve on the Internet, and what would happen if DNS did not exist?

Part B:
A new device connects to a network and starts communicating with the Internet.
Draw a diagram showing the device, router, and Internet.
Answer these questions based on the diagram:

- How the device gets an IP address
- How private networks access the Internet
- Role of the router in communication

Total Marks: 50
Total Time: 1 hour and 45 minutes